

JOSHUA SLAUGHTER

 GitHub  Google Scholar  LinkedIn

joshua.slaughter@ed.ac.uk

EDUCATION

University of Edinburgh

Edinburgh, United Kingdom

School of Informatics — Human Genetics Unit, *Marshall Scholar*

Ph.D. in Informatics

October, 2026

Advisors: Ava Khamseh and Chris Ponting

Committee: Ajitha Rajan and Caroline Dalton

University of Maryland, Baltimore County

Baltimore, MD, USA

Meyerhoff Scholar and Goldwater Scholar

B.S. Computer Engineering

August 2018 – May 2022

Magna Cum Laude

GPA: 3.8/4.0

RELEVANT COURSEWORK

Methods for Causal Inference, Targeted Causal Learning, Bias and Justice in Tech, Artificial Intelligence, Machine Learning, Signal Processing for Big Data, Probability and Statistics, Real Analysis, Linear Algebra, Differential Equations

RESEARCH INTERESTS

Causal Inference, Targeted Learning, Health Data Science, Statistical Genetics, Social Policy, Experimental Design, Machine Learning, Signal Processing

TECHNICAL SKILLS

Programming	Julia, Python, R, Bash, SQL, MATLAB
Bioinformatics	Nextflow, PLINK, GCTA
Machine Learning	TensorFlow, PyTorch, scikit-learn, MLJ
DevOps & Tools	Git, Github Actions, Docker, Singularity, Conda, Linux/HPC environments
Scientific Writing	L ^A T _E X, Overleaf, R Markdown, Microsoft Office
Other Software	ArcGIS, COMSOL Multiphysics

RESEARCH EXPERIENCE

University of Edinburgh

October 2022 – Present

Ph.D. Researcher – Advisors: Ava Khamseh and Chris Ponting

- Developed *TarGene*, a bioinformatics pipeline for semi-parametric estimation of genetic effects on complex traits at genome-wide scale.
- Implemented semi-parametric estimation strategies in the Julia module, *TMLE.jl* for population parameter estimation under biased sampling regimes.
- Uncovered and replicated novel genetic locus associated with ME/CFS risk across independent cohorts.
- Provided project design guidance and support for DecodeME, the world's largest ME/CFS study.

- Performed functional analyses to link ME/CFS-associated loci to regulatory functions elucidating disease mechanisms.
- Continued to maintain open-source machine learning software in *TMLE.jl* and *TarGene*, utilising best practices for testing and software development including building continuous delivery/continuous deployment (CI/CD) pipelines for automated testing and deployment.

Princeton Neuroscience Institute

May 2020 – August 2021

Undergraduate Researcher – Advisors: Jonathan Cohen and Marius Cătălin Iordan

- Extracted multi-layer feature activations from pretrained convolutional neural networks and correlated them with neuronal activity in the visual stream.

University of Maryland, Baltimore County

February 2019 – May 2022

Undergraduate Researcher – Machine Learning & Signal Processing Laboratory

Advisors: Tülay Adalı and Zois Boukouvalas

- Developed data-driven methods to identify fMRI biomarkers for classifying schizophrenia subtypes.
- Applied independent component analysis and machine learning techniques to neuroimaging data.

Carnegie Mellon University

May 2019 – August 2019

Undergraduate Researcher – REUSE Program, Institute for Software Research

Advisors: Justine Sherry and Srinivasan Seshan

- Developed classification methods to differentiate TCP congestion control algorithms using non-congestive loss patterns.

University of Maryland, Baltimore County

June 2017 – May 2019

Undergraduate Researcher – The Earth From Above Laboratory

Advisor: Matthew Fagan

- Analyzed tree plantation expansion and forest fates across tropical biomes using remote sensing data.

VOLUNTEER EXPERIENCE

ForKids

June 2026 – Present

Health Policy Analyst – Research, Advocacy, and Government Relations

Supervisor: Bjorn Koxvold

- Analyzed benefits cliff effects in means-standard services.
- Wrote one-pagers and policy briefs for Virginia state legislators on implementing interventions to reduce child health care uninsurance rates.
- Started a focus group consisting of professors, legislators, Virginia state department leaders, and health policy experts to address child health policy issues that the Commonwealth is currently facing.

PUBLICATIONS

Peer-Reviewed Journal Articles

- [1] Labayle O, **Slaughter J**, Roskams-Hieter B, Tetley-Campbell K, van der Laan M, Ponting CP, Khamseh A, Beentjes SV. TarGene: A Nextflow pipeline for the estimation of genetic effects on human traits via semi-parametric methods. *Journal of Open Source Software*.
<https://doi.org/10.21105/joss.09603>

- [2] Labayle O, Roskams-Hieter B, **Slaughter J**, Tetley-Campbell K, van der Laan M, Ponting CP, Beentjes SV, Khamseh A. Semi-parametric efficient estimation of small genetic effects in large-scale population cohorts. *Biostatistics*. <https://doi.org/10.1093/biostatistics/kxaf030>.
- [3] Fagan ME, Kim D, Settle W, Ferry L, Drew J, Carlson H, **Slaughter J**, Schaferbien J, Tyukavina A, Harris NL, Goldman E, Ordway EM. The expansion of tree plantations across tropical biomes. *Nature Sustainability*. 2022. [doi:10.1038/s41893-022-00904-w](https://doi.org/10.1038/s41893-022-00904-w)
- [4] Reid JL, Fagan ME, Lucas J, **Slaughter J**, Zahawi RA. The ephemerality of secondary forests in southern Costa Rica. *Conservation Letters*. 2019;12(5):e12607. [doi:10.1111/conl.12607](https://doi.org/10.1111/conl.12607)

In Preparation

- [1] **Slaughter J**, Roskams-Hieter B, Labayle O, Dibble JJ, Beentjes SV, Khamseh A, Ponting CP. Seven replicated genomic associations of myalgic encephalomyelitis/chronic fatigue syndrome: a biobank study. *medRxiv preprint*. 2026. [doi:10.64898/2026.09.09.26362115](https://doi.org/10.64898/2026.09.09.26362115)
- [2] Genetics Delivery Team, ..., **Slaughter J**, ..., Ponting CP. Initial findings from the DecodeME genome-wide association study of myalgic encephalomyelitis/chronic fatigue syndrome. *medRxiv preprint*. 2025. [doi:10.1101/2025.08.06.25333109](https://doi.org/10.1101/2025.08.06.25333109)

Conference Publications

- [3] Fagan ME, Kim D, Settle W, Ferry L, Drew J, Carlson H, Schaferbien J, **Slaughter J**. The pan-tropical expansion of tree plantations (2000-2012). *AGU Fall Meeting Abstracts*. 2020.
- [4] Fagan ME, Kim D, Ferry L, Drew J, Carlson H, Schaferbien J, **Slaughter J**. Beyond “forest” change: assessing and improving global forest cover products. *AGU Fall Meeting Abstracts*. 2018.

CONFERENCE PRESENTATIONS

- [1] **Slaughter J**, Roskams-Hieter B, Labayle O, Samms GL, Beentjes SV, Khamseh A, Ponting CP. “Targeted Learning Reveals Novel Genomic Risk Loci in ME/CFS.” Festival of Genomics. Hinxton, UK. January 2025.
- [2] **Slaughter J**, Roskams-Hieter B, Labayle O, Samms GL, Beentjes SV, Khamseh A, Ponting CP. “Targeted Learning Reveals Novel Genomic Risk Loci in ME/CFS.” Women’s Health: Genes, Data and Advancing Approaches. Hinxton, UK. January 2025.
- [3] **Slaughter J**. “Targeted Learning-Based GWAS Pipeline Towards Identifying Non-Linear Genetic Effects.” Edinburgh Next Generation of Genomics Symposium. Edinburgh, UK. October 2024.
- [4] **Slaughter J**. “Estimating Genotype and GxE Effects on Poverty-Related Diseases using Targeted Learning.” University of Edinburgh Biomedical Genomics Section Meeting. Edinburgh, UK. May 2023.
- [5] **Slaughter J**, Jordan M, Cohen J, Peterson J. “Toward Using Convolutional Neural Networks to Predict Human Behavior and Neural Representations.” Leadership Alliance National Symposium. Hartford, CT. July 2020.
- [6] **Slaughter J**, Boukouvalas Z, Adali T. “Towards Detection of Misinformation Through Machine Learning and Latent Variable Methods.” Undergraduate Research & Creative Achievement Day. Baltimore, MD. April 2020.
- [7] **Slaughter J**, Ware R, Seshan S, Sherry J. “Using Non-Congestive Loss to Differentiate TCP Reno and TCP Westwood.” REUSE Poster Symposium. Pittsburgh, PA. August 2019.
- [8] **Slaughter J**. “The ephemerality of secondary forests in southern Costa Rica.” Undergraduate Research & Creative Achievement Day. Baltimore, MD. April 2019.

- [9] **Slaughter J.** “Analysis of Geospatial Factors Relating to Opiate Overdoses for the Community of Carroll County.” Capstone Trade-Show & Presentation Night. Westminster, MD. January 2018.

TEACHING EXPERIENCE

University of Edinburgh

Methods for Causal Inference	Spring 2025
Insights Through Data	Fall 2023, Fall 2024
Representing Data	Spring 2024
Ethical Data Futures	Spring 2024
Ethics and Politics of Data	Fall 2023

University of Maryland, Baltimore County

CMSC 201: Computer Science I	2019 – 2022
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HONORS, AWARDS, AND FUNDING

Fellowships and Scholarships

Marshall Scholarship	2022 – 2025
Goldwater Scholarship	2021 – 2022
MARC U-RISE Trainee	2020 – 2022
Meyerhoff Scholarship	2018 – 2022

Awards

Undergraduate Research Award, University of Maryland, Baltimore County	2019
Best Geography and Environmental Systems (GES) Poster, Undergraduate Research & Creative Achievement Day	2019

Honor Societies

Tau Beta Pi (Engineering Honors Society)	2020
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